

# Assessment of Multi-View Consistency in Automated Product Form Analysis

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## 1. Introduction

Data-driven design research requires objective numerical descriptors of product form. A short vector describing elongation, symmetry, outline irregularity, or apparent part count is a useful input for comparing competitors, tracking design evolution, or linking form to subjective evaluations such as perceived quality<sup>1</sup>. At scale, such vectors can feed learned models that predict design outcomes directly from geometry.

The natural input for such descriptors is 3D CAD geometry, which measures volume, surface area, and curvature exactly. CAD is proprietary and unavailable for most consumer products. Product imagery is abundant, uniform, and already the modality human experts use when they judge design. Recent monocular depth-estimation models further narrow the gap by producing usable depth maps from a single RGB view<sup>2</sup>.

For an image-based pipeline to serve as the basis of a form-parameter dataset, its outputs must reflect the product rather than the specific render. This paper tests that prerequisite. We run a 23-parameter, two-signal pipeline on seven consumer-electronics products rendered at two image-set sizes (16 and 180 views) and measure whether the resulting parameter profiles are reproducible across set sizes.

## 2. Related Work

Quantifying product form has an established history in design research. For physical products, the pattern has been to operationalize aesthetic principles as parameter formulas: Lu, Tang, and Wu develop an improved equilibrium measurement incorporating form-element counts<sup>1</sup>. Parallel work in 3D shape analysis uses geometric descriptors such as curvature statistics and convex-hull solidity; Pandey et al. combine symmetry, curvature, and compactness features with pairwise human preference data across five object categories, finding that interpretable descriptors carry most of the aesthetic signal<sup>3</sup>.

Closest to the premise of this paper, Su et al. showed that view-based 2D descriptors of rendered 3D objects can match or exceed native-3D shape descriptors for shape understanding, with a single view often outperforming state-of-the-art 3D methods<sup>4</sup>. The question of how few views suffice, however, is an active one; Tang et al. survey sparse-view 3D reconstruction and identify view-count trade-offs as a central challenge<sup>5</sup>. The gap we target is the extraction step for design research specifically: most product-form work assumes either manual feature measurement or the availability of dense 3D data, neither of which scales to the dataset sizes modern learning requires.

## 3. Methodology

Each input image is decomposed into two signals. A binary mask is extracted from the alpha channel or from luminance thresholding followed by a largest-contour fill. A monocular depth map is produced by Depth-Anything-V2-Small, resized to source resolution, lightly smoothed, and normalized within the mask<sup>2</sup>.

The output is a vector of 23 parameters covering symmetry (Zabrodsky's continuous measure), proportion, curvature character, visual complexity, convexity, balance, and primitive fit<sup>6</sup>. Fifteen are mask-only silhouette descriptors; eight blend mask and depth evidence through per-parameter weights. Silhouette descriptors come from contour and convex-hull operations (solidity, rectangularity, convexity index); proportion descriptors from a rotated minimum bounding rectangle and PCA on the mask (aspect ratio, elongation); symmetry from a 36-angle rotation-and-flip IoU search. Depth-blended descriptors use the depth map as a per-pixel weight when computing balance, primitive fit, and orientation. Most parameters are calibrated to fall in [0, 1]; aspect ratio and sub-band entropy retain their native scales. The individual parameters are not new; the useful contribution is that all 23 can be

<sup>1</sup> Lu, P., Tang, J., & Wu, F. (2024). Product form design and evaluation method based on improved form aesthetic formula. *Symmetry*, 16(7), 883.

<sup>2</sup> Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., & Zhao, H. (2024). Depth Anything V2. *Advances in Neural Information Processing Systems (NeurIPS)*, 37.

<sup>3</sup> Pandey, A., et al. (2025). Modeling aesthetic preferences in 3D

shapes: A large-scale paired comparison study across object categories. *arXiv:2505.12373*.

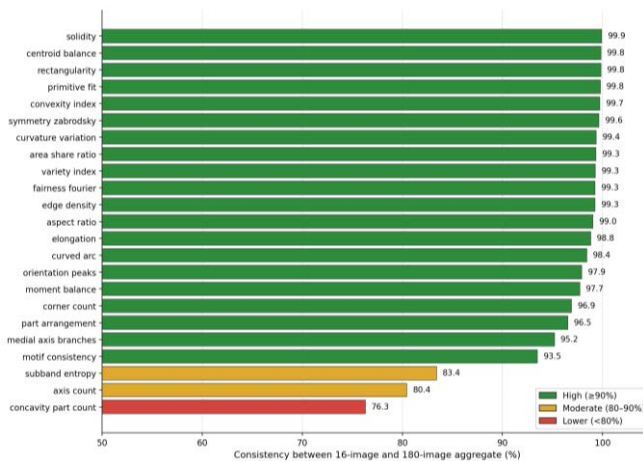
<sup>4</sup> Su, H., Maji, S., Kalogerakis, E., & Learned-Miller, E. (2015). Multi-view convolutional neural networks for 3D shape recognition. *Proceedings of ICCV*, 945–953.

<sup>5</sup> Tang, R., Naqvi, S. M. A., Yang, J., & Mountney, P. (2025). Sparse-view 3D reconstruction: Recent advances and open challenges. *arXiv:2507.16406*.

<sup>6</sup> Zabrodsky, H., Peleg, S., & Avnir, D. (1995). Symmetry as a continuous feature. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(12), 1154–1166.

obtained automatically from a small set of product images. Implementation uses OpenCV and scikit-image for classical operators and Hugging Face Transformers for the depth model.

Seven products (Lemmy, Sony PSP-1000, Teenage Engineering TP-7, a game controller, a Nokia handset, a Marshall speaker, and a Canon camera lens) were rendered through a full 360° turntable about a single vertical axis, in two set sizes: 16 images (22.5° spacing) and 180 images (2° spacing). The pipeline was applied per image, per-set aggregates were computed as arithmetic means, and the two aggregates per product were compared parameter by parameter.



[Figure 1] Per-parameter consistency averaged across 7 products

## 4. Results

Per-parameter consistency, defined as  $1 - |v_{16} - v_{180}| / |v_{180}|$  averaged across the seven products, splits the 23 parameters into three tiers (Figure 1). Twenty are highly consistent (≥90%), led by solidity (99.9%), centroid balance (99.8%), rectangularity (99.8%), primitive fit (99.8%), convexity index (99.7%), and symmetry (99.6%). Two are moderate (80–90%): subband entropy (83.4%) and axis count (80.4%). One is lower (<80%): concavity part count (76.3%).

The lower-tier parameters are integer-count or near-zero-magnitude descriptors. Subband entropy's native scale ( $\approx 10^{-3}$ ) makes small absolute deviations register as large relative ones; concavity part count and axis count are threshold-sensitive integers that drop to zero consistency on products with near-zero feature counts (concavity-part-count is 0% for PSP, 100% for Nokia and Canon Lens). They remain informative in aggregate but are lower-confidence on individual views.

Consistency is product dependent. Axis count consistency is 4.3% for TP-7 but 100% for Nokia and Canon Lens; motif consistency is 72% for PSP but ≥99% for four of the seven. Consistency profiles themselves carry information about form complexity.

## 5. Discussion

Within the tested range, 16- and 180-view aggregates produce essentially the same parameter vector for every

product. If a dense view budget were required to characterize a product's form parameters stably, this result would not hold; for the class of analyses these 23 parameters support, sparse-view imagery is sufficient, and the additional cost of dense capture or 3D reconstruction is not recovered by better parameter values.

The practical consequence is throughput. A 23-parameter score can be produced in seconds from sixteen renders, which are either already available in product catalogs or can be generated quickly. At the scale of hundreds or thousands of products, this enables paired datasets of (form parameters, expert evaluation) at the size learning models need, and category-scale comparative analysis using marketing imagery alone.

Three limitations qualify the result. The sample is seven products rendered about a single vertical axis; yaw is fully covered (360°) but pitch is not, and findings should be revisited at twenty or more products with pitch variation. The integer-count parameters are lower-confidence on individual views and need either rescaling or absolute-tolerance criteria before they can be reported on equal footing with the calibrated [0, 1] descriptors. The study establishes view-count stability, not external geometric accuracy; downstream work connecting these vectors to expert evaluations will test whether the stability is accompanied by enough discriminative power to support design-quality prediction.

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